

Instruction: There are two options: option-1 is a coding/implementing task; and option-2 is a non-coding/application task. Choose any one and register your task immediately. Option-1 can be carried out in groups of two, but option-2 individually. Results should be presented both orally (solver demonstration) and as a compact report (max. 5 pages). Late submissions carry penalty. Plagiarism check will be performed on all reports/codes!

1 Option-1: coding task

In this assignment, you should write a computer program (Matlab is recommended) to numerically solve a fully-developed turbulent channel flow using a turbulence model (choose one from the list below). Since the mean flow is one-dimensional (1D), the transport equations can be simplified into two or three coupled 1D-diffusion equations for the velocity and one or two turbulence quantities, respectively (i.e. depending on 1- or 2-equation model). It is suggested that you use Finite-Volume method.

The governing equations in a k - based eddy viscosity model (EVM) for a *stationary* ($\frac{\partial}{\partial t} < >= 0$), *spanwise homogeneous* ($\frac{\partial}{\partial z} < >= 0$), incompressible flow read [compare Eqs. 3.10, 3.12a, 3.12b, 3.35, 3.36 (Versteeg & Malalasekara [1])]

$$\begin{aligned} \frac{\partial U}{\partial x} + \frac{\partial V}{\partial y} &= 0 \\ U \frac{\partial U}{\partial x} + V \frac{\partial U}{\partial y} &= -\frac{1}{\rho} \frac{\partial P}{\partial x} + \frac{\partial}{\partial x} \left[(\nu + \nu_t) \frac{\partial U}{\partial x} \right] + \frac{\partial}{\partial y} \left[(\nu + \nu_t) \frac{\partial U}{\partial y} \right] \\ U \frac{\partial V}{\partial x} + V \frac{\partial V}{\partial y} &= -\frac{1}{\rho} \frac{\partial P}{\partial y} + \frac{\partial}{\partial x} \left[(\nu + \nu_t) \frac{\partial V}{\partial x} \right] + \frac{\partial}{\partial y} \left[(\nu + \nu_t) \frac{\partial V}{\partial y} \right] \\ U \frac{\partial k}{\partial x} + V \frac{\partial k}{\partial y} &= \frac{\partial}{\partial x} \left[\left(\nu + \frac{\nu_t}{\sigma_k} \right) \frac{\partial k}{\partial x} \right] + \frac{\partial}{\partial y} \left[\left(\nu + \frac{\nu_t}{\sigma_k} \right) \frac{\partial k}{\partial y} \right] + P_k - \varepsilon \end{aligned} \quad (1)$$

Since the flow is fully developed we have $V = \partial U / \partial x = \partial k / \partial x = 0$. Hence, the above equations can be simplified as

$$0 = -\frac{1}{\rho} \frac{\partial P}{\partial x} + \frac{\partial}{\partial y} \left[(\nu + \nu_t) \frac{\partial U}{\partial y} \right] \quad (2)$$

$$0 = \frac{\partial}{\partial y} \left[\left(\nu + \frac{\nu_t}{\sigma_k} \right) \frac{\partial k}{\partial y} \right] + P_k - \varepsilon \quad (3)$$

We find that the equations reduce to simple 1D diffusion equations with some complicated source terms.

The horizontal channel has a height of $y_{max} = 2\delta$ between the bottom flat plate (at $y = 0$) and the top flat plate (at $y = y_{max}$). The flow is driven by the pressure gradient $\partial P / \partial x$, as shown in the above momentum equation. Since the channel flow is fully developed, $\partial P / \partial x$ should be a constant and be balanced by the wall shear stress.

For comparison of your results with the DNS data [2], the task is to compute a fully developed channel flow with $Re_\tau = u_\tau \delta / \nu = 395$. Here $u_\tau = \sqrt{\tau_w / \rho}$ is the friction velocity, where $\tau_w = \mu \frac{\partial U}{\partial y} |_{wall}$. If $\delta = 1$ (i.e. $y_{max} = 2$), we have $-\frac{\partial P}{\partial x} = \tau_w$ (**show this in your report**). Further, if $\rho = u_\tau = 1$ we get $-\frac{\partial P}{\partial x} = 1$.

Choose one model from the list below (and register your group name against it).

1.1 Prandtl's 1-equation model

Use equation 2, 3 and 4 (where $\varepsilon = C_\mu k^{3/2}/\ell_T$, $C_\mu = 0.07 - 0.09$, $\sigma_k = 1$, ℓ_T is user-defined)

$$\nu_t = C_\mu \frac{k^2}{\varepsilon} \quad P_k = \nu_t \left(\frac{\partial U}{\partial y} \right)^2 \quad (4)$$

1.2 Standard $k - \varepsilon$

Use equation 2, 3, 4 and 5 (where $C_\mu = 0.09$, $C_1 = 1.44$, $C_2 = 1.92$, $\sigma_\varepsilon = 1.3$, $\sigma_k = 1$).

$$0 = \frac{\partial}{\partial y} \left[\left(\nu + \frac{\nu_t}{\sigma_\varepsilon} \right) \frac{\partial \varepsilon}{\partial y} \right] + \frac{\varepsilon}{k} (C_1 P_k - C_2 \varepsilon) \quad (5)$$

1.3 RNG $k - \varepsilon$

Use equation 2, 3, 4 and 5 (where $C_\mu = 0.085$, $\sigma_k = 1$, $\sigma_\varepsilon = 0.719$, $C_2 = 1.68$).

$$C_1 = 1.42 - \frac{\eta(1 - \eta/\eta_o)}{1 + \beta\eta^3} \quad (6)$$

where $\eta = Sk/\varepsilon$, $\eta_o = 4.38$, $\beta = 0.012$ and $S = \sqrt{2S_{ij}S_{ij}} = \sqrt{(\partial U/\partial y)^2}$

1.4 Wilcox's $k - \omega$

Use equation 2, 7, 8 and 9 (where $\beta^* = 9/100$, $\sigma_k = 2$, $\sigma_\omega = 2$, $C_1 = 5/9$, $C_2 = 3/40$).

$$0 = \frac{\partial}{\partial y} \left[\left(\nu + \frac{\nu_t}{\sigma_k} \right) \frac{\partial k}{\partial y} \right] + P_k - \beta^* \omega k \quad (7)$$

$$0 = \frac{\partial}{\partial y} \left[\left(\nu + \frac{\nu_t}{\sigma_\omega} \right) \frac{\partial \omega}{\partial y} \right] + \frac{\omega}{k} C_1 P_k - C_2 \omega^2 \quad (8)$$

$$\nu_t = \frac{k}{\omega} \quad P_k = \nu_t \left(\frac{\partial U}{\partial y} \right)^2 \quad (9)$$

Boundary conditions

The mean flow is symmetric about the channel centreline. As a consequence, you may take only a half of the channel as your computation domain (from wall to centreline). **Give a clear description in the report about the boundary conditions you have implemented.** At the wall, the boundary conditions are

$U = k = 0$	$\partial \varepsilon / \partial y = 0$ or $\varepsilon_{y^+ \leq 3} = 2\nu k / y^2$ or $\varepsilon_{y^+ \leq 3} = 2\nu (\partial \sqrt{k} / \partial y)^2$	$\omega_{y^+ \leq 3} = 6\nu / y^2 C_2$
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Numerical tips

While implementing the source terms, you may split it into S_U and S_P as

$$S = S_P \Phi_P + S_U \quad (10)$$

where $S_P < 0$. It is very important that the turbulence quantities stay positive during the iteration process. For example, if ε goes negative then ν_t becomes negative, and usually this leads to rapid divergence. Also,

if k becomes negative, the production term in the ε equation will be negative, which also will cause convergence problems. The remedy is to make sure that *all* negative source terms are put in S_P (see [1] for more information). Solve the equation system using an iterative method, e.g. Gauss-Seidel.

2 Option-2: application task

Here, you should solve a fully-developed turbulent channel flow (same problem as in Option-1) using a commercial- or open-source software (e.g. OpenFoam, Fluent, CFX, StarCCM+).

The horizontal channel has a height of $y_{max} = 2\delta$ between the bottom flat plate (at $y = 0$) and the top flat plate (at $y = y_{max}$). The flow is driven by the pressure gradient $\partial P/\partial x$. Since the channel flow is fully developed, $\partial P/\partial x$ should be a constant and be balanced by the wall shear stress.

For comparison of your results with the DNS data [2], the task is to compute a fully developed channel flow with $Re_\tau = u_\tau \delta / \nu = 395$. Here $u_\tau = \sqrt{\tau_w / \rho}$ is the friction velocity, where $\tau_w = \mu \frac{\partial U}{\partial y}|_{wall}$. If $\delta = 1$ (i.e. $y_{max} = 2$), we have $-\frac{\partial P}{\partial x} = \tau_w$ (**show this in your report**). Further, if $\rho = u_\tau = 1$ we get $-\frac{\partial P}{\partial x} = 1$.

The mean flow is symmetric about the channel centreline. As a consequence, you may take only a half of the channel as your computation domain (from wall to centreline). **Give a clear description in the report about the boundary conditions you have used. Show grid convergence.**

- i. Choose a minimum of one eddy viscosity model (EVM) from each of the family, i.e. $k - \varepsilon$ (Standard, RNG or Realizable), $k - \omega$ (Standard or SST), k-L ($k - kL$, $k - \sqrt{k}L$ or $k - kL - \omega$), v2f ($k - \varepsilon - \overline{v'^2} - f$ or $k - \varepsilon - \zeta - f$).
- ii. Among the four chosen models, you must try both LRN and wall-function formulation in at least one model.

3 Results: solver demonstration and report (max. 5 pages)

The following points must be covered in your simulation and report:

- Compare your results $U, k, \varepsilon, -\overline{uv}, \nu_t$ (and other modeled quantities) with DNS data [2]. Analyse results close to the wall.
- Study the budget of your k equation, i.e. plot the different terms (production, diffusion, etc.), and compare with the DNS data [2].

In the `dns-data.dat` file, the six columns correspond to (all scaled by u_τ and ν)

1. y^+
2. Dissipation ε
3. Production $-\overline{uv} \frac{\partial U}{\partial y}$
4. Diffusion due to pressure-velocity fluctuations $-\frac{\partial(\overline{pv})/\rho}{\partial y}$
5. Turbulent diffusion $-\frac{\partial \overline{vk'}}{\partial y}$
6. Viscous diffusion $\nu \frac{\partial^2 k}{\partial y^2}$

References

- [1] H. Versteeg and W. Malalasekera. *An Introduction to Computational Fluid Dynamics - The Finite Volume Method*. Longman Scientific & Technical, Harlow, England, 1st edition, 1995.
- [2] R. D. Moser, J. Kim, and N. N. Mansour. Direct numerical simulation of turbulent channel flow up to $Re_\tau=590$. *Physics of Fluids*, 11(4):943–945, 1999.